A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Organic nitrogen compounds

Nitrogen's lone pair is the key to this topic. It makes amines basic, lets them act as ligands and nucleophiles, and joins monomers into nylon, Kevlar and proteins. You will also meet azo dyes, the zwitterions of amino acids, and the difference between addition and condensation polymers.

SPEC 19.1 Naming and drawing amines, amides and amino acids

WHAT THIS SPEC POINT ASKS

Name amides, amines and amino acids, and draw their structural, displayed and skeletal formulae.

Amines

- An amine is ammonia with one or more H atoms replaced by carbon groups.
- Classify by the **number of carbon groups on the nitrogen** (not by the carbon the N is on, as you do for alcohols):

Type	General formula	Example	Name
Primary (1°)	RNH_2	$CH_3CH_2CH_2CH_2NH_2$	butylamine (butan-1-amine)
Secondary (2°)	R_2NH	$(CH_3)_2NH$	dimethylamine
Tertiary (3°)	R_3N	$(CH_3)_3N$	trimethylamine
Quaternary ammonium ion	R_4N^+	$(CH_3)_4N^+$	tetramethylammonium ion

- **Naming**: either name the alkyl group + "amine" (methylamine, ethylamine, propylamine), or use the IUPAC suffix "-amine" with a locant: **propan-1-amine**, **propan-2-amine**, **butan-1-amine**.
- Groups on the N in secondary and tertiary amines get the locant **N**: $CH_3NHCH_2CH_3$ is **N-methylethylamine** (or N-methylethanamine).
- When a higher-priority group is present (e.g. –COOH), the NH_2 becomes the prefix **amino-**: 2-aminopropanoic acid.
- **Aliphatic** amines have N joined to an alkyl chain. **Aromatic** amines have N joined directly to a benzene ring: $C_6H_5NH_2$ is **phenylamine** (also called aniline or aminobenzene).

Amides

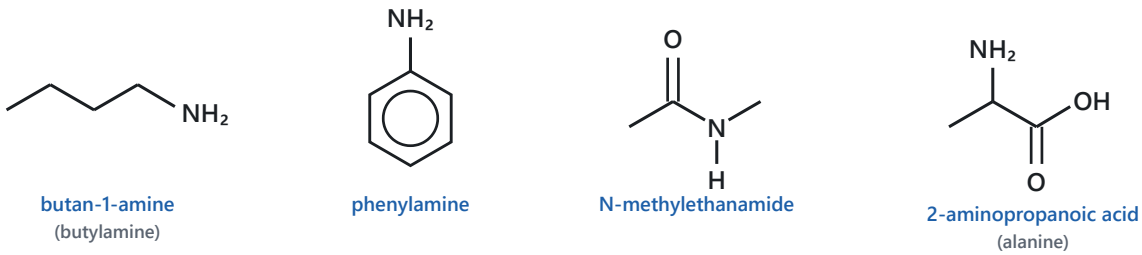
- Amides contain **–CONH₂** (or –CONHR, –CONR₂). The C=O carbon counts as carbon 1 of the chain.
- Name: stem + **"-amide"**. CH_3CONH_2 is **ethanamide**, $CH_3CH_2CONH_2$ is **propanamide**, $C_6H_5CONH_2$ is **benzamide**, $CH_2{=}CHCONH_2$ is **prop-2-enamide** (propenamide).
- Groups on the nitrogen use the **N-** locant: $CH_3CONHCH_3$ is **N-methylethanamide**; $CH_3CONHCH_2CH_2CH_2CH_3$ is **N-butylethanamide**; $CH_3CH_2CON(CH_3)_2$ is **N,N-dimethylpropanamide**. Two different groups go in alphabetical order (N-ethyl-N-methyl…).

Amino acids

- An amino acid has an **amine group (–NH₂)** and a **carboxylic acid group (–COOH)**.
- The natural ones are **2-amino acids**: both groups are on carbon 2 (the carbon next to COOH). General formula **RCH(NH₂)COOH**. About 20 of them make up proteins, differing only in the **R group**.
- Name them as substituted carboxylic acids (the acid takes priority):

Common name	R group	Systematic name
glycine	H	2-aminoethanoic acid (aminoethanoic acid)
alanine	CH_3	2-aminopropanoic acid
serine	CH_2OH	2-amino-3-hydroxypropanoic acid
phenylalanine	$CH_2C_6H_5$	2-amino-3-phenylpropanoic acid
aspartic acid	CH_2COOH	2-aminobutanedioic acid

Drawing them



Skeletal formulae. Hydrogens on N and O are always drawn; hydrogens on C are not.

- **Structural**: $CH_3CH(NH_2)COOH$. **Displayed**: every atom and every bond drawn, including N–H and O–H. **Skeletal**: carbon chain as zigzag lines, with N, O and the **H atoms on N and O** written in.

- To get a **molecular formula from a skeletal formula**, count the carbons at each line end and corner. Then give each carbon enough H atoms to make four bonds.

MARK-SCHEME WORDING

- Classify: "tertiary (amine) / 3°". "3rd / third" and "tertiary **amide**" are not accepted (WCH15 Jan 2022, Jun 2025).
- Names credited: "2-methylbutanamide" (WCH15 Jan 2021), "2-amino-3-methylpentanoic acid" (WCH15 Jun 2022), "2-amino-3-hydroxypropanoic acid" (WCH15 Oct 2024), "N-butylethanamide" (WCH15 Jun 2025), "prop-2-enamide / propenamide" (not "propanamide") (WCH15 Jun 2023).
- Functional groups in a drug molecule: "ester", "(primary) amine", "arene". Not "ketone" or "carboxylic acid" for an ester. "Carbonyl" is kept for aldehydes and ketones only.

Often asked as: name a compound from its skeletal formula; classify an amine; give the molecular formula from a skeletal one; identify functional groups.

COMMON MISTAKE

Classifying amines like alcohols (by the carbon). Count the groups on the **N**.

In skeletal formulae, leaving the H off N or O (–N– instead of –NH–), or an O–H joined to the wrong atom.

Naming an amino acid "aminopropanoic-2 acid" or putting the amino group last. The acid gives the suffix; amino is a prefix with a number.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

IUPAC names of other multifunctional compounds turn up in the same questions: **2-hydroxybenzoic acid** (salicylic acid, WCH15 Oct 2023) and **propane-1,2,3-triol** (glycerol, WCH15 Jun 2024). The suffix names the main group, and everything else is a numbered prefix.

SPEC 19.2 Reactions of primary amines

WHAT THIS SPEC POINT ASKS

Understand the reactions of butylamine (aliphatic) and phenylamine (aromatic) with: (i) water, giving an alkaline solution; (ii) acids, giving salts; (iii) halogenoalkanes; (iv) ethanoyl chloride; (v) copper(II) ions, giving a complex ion.

The key is the **lone pair on the nitrogen**. It lets an amine act as a **base** (accepts H^+), a **nucleophile** (attacks δ+ carbon) and a **ligand** (bonds to a metal ion).

(i) With water: an alkaline solution

$$CH_3CH_2CH_2CH_2NH_2(aq) + H_2O(l) \rightleftharpoons CH_3CH_2CH_2CH_2NH_3^+(aq) + OH^-(aq)$$

- The N lone pair accepts H^+ from a water molecule, leaving **OH^- ions**, so the solution is **alkaline** (about pH 11–12). Butylamine is a **weak base**: the equilibrium lies to the left.
- Phenylamine is only slightly soluble and a much weaker base. Its solution is only **very weakly alkaline** (19.3).

(ii) With acids: salts

$$CH_3CH_2CH_2CH_2NH_2 + HCl \rightarrow CH_3CH_2CH_2CH_2NH_3^+Cl^- \text{ (butylammonium chloride)}$$
$$C_6H_5NH_2 + HCl \rightarrow C_6H_5NH_3^+Cl^- \text{ (phenylammonium chloride)}$$

- The salts are **ionic, white crystalline solids** that are **soluble in water**. Insoluble phenylamine **dissolves** in hydrochloric acid because it turns into the soluble salt.
- The H^+ joins the N by a **dative covalent bond** using the lone pair.
- Adding **NaOH(aq)** gets the amine back: $C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$. This step is used in 19.5.

(iii) With halogenoalkanes: further alkylation

$$CH_3CH_2CH_2CH_2NH_2 + CH_3CH_2Br \rightarrow CH_3CH_2CH_2CH_2NHCH_2CH_3 + HBr$$

- **Nucleophilic substitution**: the N lone pair attacks the δ+ carbon of C–Br (the same mechanism as NH_3 in Unit 2). The HBr formed reacts with more amine to give a salt.
- The product is itself an amine with a lone pair, so it can react again: **primary → secondary → tertiary amine → quaternary ammonium salt**. You get a **mixture**.
- With **excess halogenoalkane** the main product is the **quaternary ammonium salt**, e.g. $(C_4H_9)(C_2H_5)_3N^+Br^-$. Phenylamine reacts the same way, but more slowly.

(iv) With ethanoyl chloride: N-substituted amides

$$CH_3COCl + CH_3CH_2CH_2CH_2NH_2 \rightarrow CH_3CONHCH_2CH_2CH_2CH_3 + HCl \text{ (**N-butylethanamide**)}$$
$$CH_3COCl + C_6H_5NH_2 \rightarrow CH_3CONHC_6H_5 + HCl \text{ (**N-phenylethanamide**)}$$

- Vigorous at room temperature; **steamy fumes of HCl**. With excess amine, the HCl is taken up as a salt: $CH_3COCl + 2RNH_2 \rightarrow CH_3CONHR + RNH_3^+Cl^-$.
- **How it happens (addition–elimination)**:
  1. The carbonyl carbon is **δ+**: both O and Cl pull electrons away from it.
  2. The **lone pair on N** attacks this carbon, and the C=O π electrons move onto the O.
  3. The C=O re-forms and the **C–Cl bond breaks**, releasing Cl^- (a good leaving group).
  4. H^+ is lost from the N, giving the amide and HCl.

(v) With copper(II) ions: a complex

- Add butylamine solution drop by drop to $CuSO_4(aq)$:
  - A little amine: **pale blue precipitate** of copper(II) hydroxide. The amine acts as a base, making OH^-: $[Cu(H_2O)_6]^{2+} + 2RNH_2 \rightarrow [Cu(OH)_2(H_2O)_4] + 2RNH_3^+$
  - Excess amine: the precipitate **dissolves** to a **deep blue solution**. The amine acts as a ligand: $[Cu(H_2O)_6]^{2+} + 4RNH_2 \rightarrow [Cu(RNH_2)_4(H_2O)_2]^{2+} + 4H_2O$
- This is just like ammonia (Topic 17): N lone pairs form dative bonds to Cu^{2+}.
- With **phenylamine** the expected observation is the **pale blue precipitate** (WCH15 Oct 2020).

MARK-SCHEME WORDING

- Amine in water: "the nitrogen has a lone pair", "which can accept a H^+ (from water)", "leaving (an excess of) hydroxide ions" (WCH15 Jan 2022, pyridine).
- Butylamine + ethanoyl chloride: the **displayed** formula of N-butylethanamide **and HCl** (WCH15 Oct 2024).
- Amide-formation explanation, any three: "carbonyl carbon is electron-deficient / δ+", "nitrogen has a lone pair", "which forms a bond to the carbonyl carbon", "C–Cl bond breaks / Cl is a good leaving group" (WCH15 Jun 2022). The name of the mechanism earns nothing on its own.
- Halogenoalkane mechanism (WCH15 Jan 2022): "arrow from lone pair on N to C", "dipole and arrow from C–Cl to Cl", "intermediate with + on N", "arrow from N–H bond to N^+". N^- is never awarded.
- Cu^{2+} + phenylamine: "(pale) blue precipitate"; blue-green is not accepted.
- Salt with HCl: "procaine hydrochloride is ionic", "the ions are hydrated by water", so it is more soluble (WCH15 Jan 2025).

Often asked as: write the equation (displayed or molecular formulae), explain why a solution is alkaline, give observations with Cu^{2+}, explain why a drug is sold as its hydrochloride.

COMMON MISTAKE

Leaving out the HCl in the ethanoyl chloride equation (Oct 2024 report).

Writing that the amine attacks a carbocation. In the acyl chloride reaction there is no carbocation: the N attacks the δ+ carbon of C=O (Jun 2022 report).

Mixing up the words base (takes H^+) and nucleophile (attacks δ+ carbon). Both use the lone pair, but they are different roles.

Putting the H^+ on the wrong N in a molecule with two nitrogens, or drawing the N–H^+ bond without a dative arrow when the question asks for one.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Tertiary and ring amines** (pyridine, trimethylamine) behave the same way: base in water, nucleophile towards chloromethane (WCH15 Jan 2022, Jun 2025).
- **Drugs as salts**: amines are converted into hydrochlorides because the ionic salts are **more soluble in water** (ion–dipole attractions with water) (WCH15 Jan 2025).

SPEC 19.3 Solubility of amines and their strength as bases

WHAT THIS SPEC POINT ASKS

Understand that amines mix with water because of hydrogen bonding, and explain why ammonia, primary aliphatic amines and primary aromatic amines differ in base strength.

Mixing with water

- Small amines (methylamine to butylamine) are **miscible** with water.
- Two kinds of **hydrogen bond** form:
  - the **lone pair on the amine N** with a δ+ H of water: N···H–O;
  - a δ+ **H on the amine N** with a lone pair on water's O: N–H···O.
- As the hydrocarbon chain gets longer, the non-polar part disrupts the hydrogen bonding, so **solubility falls**. Phenylamine, with its large benzene ring, is only **slightly soluble**.
- Primary amines boil higher than alkanes of similar M_r (they have hydrogen bonds between molecules), but lower than alcohols. N is less electronegative than O, so N–H···N hydrogen bonds are weaker than O–H···O.

EXAM TIP

Hydrogen-bond diagrams: draw the **lone pair** on the N (or O) that accepts the bond, label δ+ on H and δ− on N and O, and keep N···H–O in a **straight line** (about 180°). Use a dashed line for the hydrogen bond, never a solid line or a dative arrow.

Base strength

A base **accepts a proton (H^+)**, using the **lone pair on nitrogen**. The more available that lone pair is, the stronger the base.

Base	pK_a of its conjugate acid	Strength	Reason
butylamine	about 10.6	strongest	alkyl group **releases electrons** (positive inductive effect), raising the electron density on N, so the lone pair is **more available**

Base	pK_a of its conjugate acid	Strength	Reason
ammonia	9.3	in between	reference: only H atoms on N
phenylamine	about 4.6	weakest	lone pair on N **overlaps with the delocalised π system** of the ring, so electron density on N is lower and the lone pair is **less available**

Order: butylamine > ammonia > phenylamine. (A higher pK_a of the conjugate acid means a stronger base. You don't need to learn these numbers.)

KEY IDEA

Model answer (why phenylamine is a weaker base than ammonia):

1. Bases accept H^+ using the lone pair on nitrogen.
2. In phenylamine, the **lone pair on N overlaps with (is partly delocalised into) the π electrons of the benzene ring**.
3. So the lone pair is **less available** and nitrogen is **less able to accept H^+** than in ammonia.

MARK-SCHEME WORDING

- Phenylamine: "lone pair on N overlaps with the π cloud / delocalised electrons / interacts with the ring", "so the N is less able to accept a H^+ / proton" (WCH15 Oct 2020). "Lone pair is less available to accept H^+ / form a dative bond" is accepted.
- Butylamine: "acts as a base by accepting a proton", "greater electron density on the nitrogen", "as a result of the alkyl group releasing electron density" (WCH15 Oct 2024). "Positive inductive effect" and "electron pushing" are accepted. "Nitrogen is more negative" and electronegativity arguments are not.
- Hydrogen bond: "lone pair (of electrons)" included; "dipoles on N, H and O"; "linear about the H" (170–190°) (WCH15 Oct 2024, Jun 2025).

Often asked as: rank three bases (MCQ), explain the order (2–3 marks), draw a hydrogen bond between an amine and water.

COMMON MISTAKE

Finishing at "the lone pair is delocalised" without the conclusion: **less able to accept H^+** (Oct 2020 report).

Saying the benzene ring makes the aromatic amine **more** likely to be protonated: the argument goes the other way (Jun 2023 report).

Not defining basicity. The first mark is often just "a base accepts a proton / H^+" (Oct 2024 report).

Hydrogen-bond diagrams with no lone pair, or with the bond bent.

⚠ Beyond the spec · not part of the Edexcel IAL specification

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- **Two nitrogens in one molecule** (e.g. the anaesthetic procaine): the tertiary **alkyl** amine N is the more basic one, and the **aromatic** amine N, whose lone pair is drawn into the ring, is less basic. Discuss **both** nitrogens (WCH15 Jun 2023, Jan 2025).
- **Amides are not basic**: the N lone pair is delocalised onto the C=O group.

SPEC 19.4 Making primary aliphatic amines

WHAT THIS SPEC POINT ASKS

Know the reagents and general conditions for making primary aliphatic amines (i) from halogenoalkanes and (ii) by reducing nitriles.

(i) From a halogenoalkane and ammonia

$$CH_3CH_2CH_2CH_2Br + 2NH_3 \rightarrow CH_3CH_2CH_2CH_2NH_2 + NH_4Br$$

- **Reagents and conditions**: **excess concentrated ammonia in ethanol**, heated in a **sealed tube** (under pressure).
- Why each one:
  - **Sealed tube**: ammonia is a gas and would escape from an open flask when heated.
  - **Ethanol** as solvent: in water, OH^- ions would compete and give some alcohol.
  - **Excess ammonia**: the primary amine product is also a nucleophile and can react with more halogenoalkane (19.2 iii). Lots of NH_3 makes it more likely that each halogenoalkane meets NH_3 instead. Even so, you get some secondary and tertiary amine.

(ii) By reducing a nitrile

$$CH_3CH_2CH_2CN + 4[H] \rightarrow CH_3CH_2CH_2CH_2NH_2$$

- **Reagent**: **lithium tetrahydridoaluminate(III), $LiAlH_4$, in dry ether**, then dilute acid. (Catalytic hydrogenation with H_2 and a nickel catalyst also works, but one mark scheme refused it, so $LiAlH_4$ is the safe answer.)
- The nitrile comes from a halogenoalkane: **KCN in ethanol, heat under reflux**. $CH_3CH_2CH_2Br + CN^- \rightarrow CH_3CH_2CH_2CN + Br^-$
- So the nitrile route **adds one carbon** to the chain: 1-bromopropane (3 C) → butanenitrile (4 C) → butylamine (4 C).

- It also gives **only the primary amine**, with no further alkylation.

Route	Reagents	Carbon chain	Product purity
Halogenoalkane + NH_3	excess NH_3, ethanol, heat, sealed tube	same length	mixture of amines
Nitrile reduction	$LiAlH_4$ in dry ether	one longer than the halogenoalkane used	primary amine only

MARK-SCHEME WORDING

- "potassium cyanide and (aqueous) ethanol / ethanolic KCN" (NaCN accepted). KCN "in acid" or HCN for a halogenoalkane substitution is not accepted.
- "$LiAlH_4$ / lithium tetrahydridoaluminate(III) in (dry) ether". Oct 2020 also allowed H_2 + Ni or sodium and ethanol, but Jan 2023 refused H_2 + Ni/Pt/Pd.
- Nitrile **hydrolysis** (to an acid, not an amine): "aqueous acid / $H^+(aq)$" and "reflux". Concentrated acid and "oxidation" are not accepted.

Often asked as: reagents for a step in a multi-step synthesis (WCH15 Jan 2023, 5 marks: ethanolic KCN, then $LiAlH_4$, then ethanoyl chloride).

COMMON MISTAKE

Forgetting "ethanolic" for KCN (Jan 2023 report). With aqueous KCN you also get OH^- substitution.

Using a reagent that removes or adds the wrong number of carbons. Count carbons at every step.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Nitrile hydrolysis: heating a nitrile under reflux with dilute HCl gives the **carboxylic acid**: $CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$. Alkaline hydrolysis gives the carboxylate ion, which must then be acidified (WCH15 Jun 2024, Jan 2026). Together with KCN, this is a way to add one carbon and make an acid.

SPEC 19.5 Making aromatic amines: reducing nitro compounds

WHAT THIS SPEC POINT ASKS

Know that aromatic amines are made by reducing aromatic nitro compounds with tin and concentrated

hydrochloric acid.

$$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$$

Method

1. Heat **nitrobenzene** with **tin** and **concentrated hydrochloric acid** under **reflux**. Tin is the **reducing agent**: it is oxidised to tin(II)/tin(IV), and supplies the electrons. It is **not a catalyst**.
2. In the acid, the product is the **phenylammonium ion**, $C_6H_5NH_3^+$ (as a salt with chloride and tin ions).
3. Add **excess sodium hydroxide solution** to liberate the free amine: $C_6H_5NH_3^+ + OH^- \rightarrow C_6H_5NH_2 + H_2O$ (tin hydroxides also precipitate).
4. Separate the phenylamine by **steam distillation** (20.5), then purify.

- This is a **reduction** (O atoms removed, H atoms added).
- Overall route from benzene: **nitration** (conc. HNO_3/conc. H_2SO_4, 50–55 °C) → **reduction** (Sn/conc. HCl) → phenylamine.
- **Uses of phenylamine**: making **azo dyes** (19.6), and drugs and other chemicals.

MARK-SCHEME WORDING

- "tin and (concentrated) hydrochloric acid" (Fe or Zn in place of Sn accepted; "heat / reflux" ignored).
- Not accepted: dilute acid, sulfuric or nitric acid, $LiAlH_4$, "Sn is a catalyst", or NaOH **in the same step** as the acid.
- Type of reaction: "reduction" (redox accepted; "hydrogenation" ignored).

Often asked as: reagents for the nitro → amine step (most series), name the type of reaction, write the equation using [H].

COMMON MISTAKE

Only about 40% of candidates knew this reagent in one series (Jan 2022 report). Learn it: **Sn + concentrated HCl**.

Suggesting $LiAlH_4$, or calling tin a catalyst, or listing NaOH with the acid as one step (Jun 2023 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

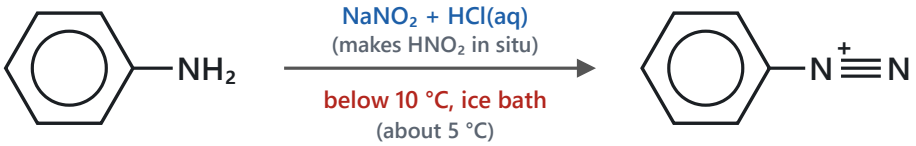
Examined in past papers even though the spec doesn't list it.

Tin and HCl reduce the NO_2 group but leave a **COOH** group alone, so 2-nitrobenzoic acid gives **2-aminobenzoic acid** (WCH15 Oct 2021). Making the mixture alkaline afterwards to free the amine is also credited (WCH15 Jun 2022 report).

SPEC 19.6 Diazotisation and azo dyes

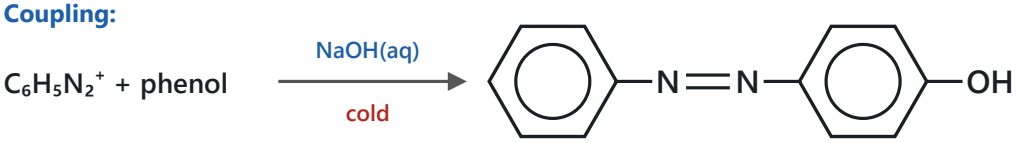
WHAT THIS SPEC POINT ASKS

Describe how an aromatic amine reacts with nitrous acid to give a benzenediazonium ion, and how this then couples with phenol to make a dye.



phenylamine — benzenediazonium ion

Coupling:



4-hydroxyazobenzene: yellow-orange dye (precipitate)
–N=N– is para (position 4) to the OH

Two steps, both kept cold: diazotisation, then coupling.

Making an azo dye from phenylamine

1. **Diazotisation**: react phenylamine with nitrous acid below 10 °C. Nitrous acid is unstable, so it is made in the flask from sodium nitrite and hydrochloric acid: $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$.

2. $C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O$, forming benzenediazonium chloride, which contains the $–N{\equiv}N^+$ group.

   Keep it cold: above about 10 °C the diazonium ion reacts with water, $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2 + H^+$, releasing nitrogen gas. Much below 0 °C the reaction is too slow.

3. **Coupling**: add the diazonium salt to phenol dissolved in sodium hydroxide. The diazonium ion is a weak electrophile that attacks the activated ring: $C_6H_5N_2^+ + C_6H_5OH \rightarrow C_6H_5–N{=}N–C_6H_4OH + H^+$.

4 The product, 4-hydroxyazobenzene, is a yellow-orange **azo dye**. The –N=N– link joins two benzene rings into one long delocalised system that absorbs visible light. Azo compounds are used as dyes, pigments and food colourings.

Step 1: diazotisation

- **Nitrous acid (HNO_2)** is unstable, so it is **made in the flask**: $NaNO_2 + HCl \rightarrow HNO_2 + NaCl$.
- Dissolve phenylamine in **excess hydrochloric acid**, cool in an **ice bath** to **below 10 °C (about 5 °C)**, and add cold **sodium nitrite (sodium nitrate(III)) solution** slowly.

$$C_6H_5NH_2 + HNO_2 + H^+ \rightarrow C_6H_5N_2^+ + 2H_2O$$

- The product is the **benzenediazonium ion**, $C_6H_5–N^+{\equiv}N$. The **+ is on the N joined to the ring**.
- **Why cold?** Above about 10 °C the diazonium ion **decomposes**: it reacts with water to give **phenol** and **nitrogen gas** (you would see bubbles).
  $C_6H_5N_2^+ + H_2O \rightarrow C_6H_5OH + N_2 + H^+$
  Below 0 °C the reaction is too slow and the solution may freeze, so don't go below 0 °C.

Step 2: coupling

- Add the cold diazonium solution to **phenol dissolved in sodium hydroxide solution** (alkaline), kept cold.
- The diazonium ion is an **electrophile**. It attacks the activated ring of phenol (as the phenoxide ion) at **position 4**, opposite the –OH: electrophilic substitution.

$$C_6H_5N_2^+ + C_6H_5OH \rightarrow C_6H_5–N=N–C_6H_4OH + H^+$$

- The product, **4-hydroxyazobenzene**, is a **yellow-orange** precipitate: an **azo dye**. The **–N=N–** link is the **azo group**.
- **Why coloured?** The N=N joins the two rings into one large **delocalised system**. This lets the molecule absorb visible light. It also makes the dye **stable**.
- Other coupling partners give other colours: naphthalen-2-ol gives a red dye; N,N-dimethylphenylamine gives methyl red (WCH15 Oct 2021).

MARK-SCHEME WORDING

- Reagents: "sodium nitrite / sodium nitrate(III) / $NaNO_2$ **and** hydrochloric acid" (HNO_2 + HCl accepted). "Sodium nitrate" and "nitric acid" are not accepted.
- Temperature: "at 5 °C / between 0 and 10 °C" (ice bath accepted). "Below 0 °C" is not accepted (WCH15 Jun 2025).
- Why: "diazonium ion decomposes / reacts with water / forms phenol" (above 10 °C); "rate too slow" (below 0 °C). "Side reactions", "yield too low" and "freezes" are ignored (WCH15 Oct 2021).
- Diazonium ion: $ArN^+{\equiv}N$, with the + on the correct N; "N_2^+" is not accepted.
- Dye: ring–N=N–ring with the OH in the right place. "–N_2–" or N≡N between the rings is not accepted.

- Coupling conditions: "phenol dissolved in an alkaline solution" at about 5 °C (WCH15 Jan 2025 MCQ).

Often asked as: reagents and temperature (2 marks, most series); draw the diazonium ion or the dye; explain the temperature; the structure of a dye from an unfamiliar amine and coupling partner.

COMMON MISTAKE

Writing **nitrate** or nitric acid instead of **nitrite** (sodium nitrate(III)).

Drawing N≡N or –N_2– in the dye. The dye has a **double** bond, N=N.

Putting the OH next to the azo link when phenol couples. It ends up **para** (position 4); with 2-hydroxybenzoic acid the positions follow the stem (Jan 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **E/Z isomerism about N=N**: there is restricted rotation about the N=N double bond, so azo dyes have E and Z isomers (WCH15 Oct 2020).
- **Sulfonate groups** (–SO_3^- Na^+) are added to dyes such as sunset yellow to make them **soluble in water** (WCH15 Jun 2025).
- Dye **purification by recrystallisation** and percentage-yield questions (WCH15 Jun 2025).

SPEC 19.7 Making amides from acyl chlorides

WHAT THIS SPEC POINT ASKS

Understand that amides can be made from acyl chlorides.

- An acyl chloride reacts with **ammonia** to give an **amide**, or with a **primary amine** to give an **N-substituted amide**. Both happen quickly at **room temperature**, giving steamy fumes of HCl.

Reactants	Equation	Product
propanoyl chloride + ammonia	$CH_3CH_2COCl + NH_3 \rightarrow CH_3CH_2CONH_2 + HCl$	propanamide
with excess ammonia	$CH_3CH_2COCl + 2NH_3 \rightarrow CH_3CH_2CONH_2 + NH_4Cl$	propanamide
propanoyl chloride + methylamine	$CH_3CH_2COCl + CH_3NH_2 \rightarrow CH_3CH_2CONHCH_3 + HCl$	N-methylpropanamide

Reactants	Equation	Product
ethanoyl chloride + phenylamine	$CH_3COCl + C_6H_5NH_2 \rightarrow CH_3CONHC_6H_5 + HCl$	N-phenylethanamide

- With excess ammonia, the HCl reacts with ammonia to form **ammonium chloride**, a white solid.
- The mechanism is **addition–elimination** (19.2 iv): N lone pair → δ+ carbonyl carbon; C–Cl breaks; H^+ lost.
- To make an amide from a **carboxylic acid**, first convert the acid to the acyl chloride with **PCl_5** (or $SOCl_2$), then add NH_3. For example, benzoic acid → benzoyl chloride → **benzamide** (WCH15 Oct 2023).
- **Ethanoic anhydride** does the same job more slowly and gives ethanoic acid instead of HCl (see the aspirin practical, 20.4).

MARK-SCHEME WORDING

- "ammonia / NH_3 added to an acyl chloride"; to get the acyl chloride, "phosphorus(V) chloride / PCl_5 (or thionyl chloride)". "Ammonia added to benzoic acid" is not accepted: it only makes the ammonium salt (WCH15 Oct 2023).
- Acylating agent: "ethanoyl chloride or ethanoic anhydride". "Ethanoic acid" is not accepted (WCH15 Jan 2023).

Often asked as: the last step of a synthesis route; the product of an acyl chloride with a given amine; the equation including HCl.

COMMON MISTAKE

Adding ammonia straight to a carboxylic acid to make an amide. That is an acid–base reaction: it gives an **ammonium salt** (Oct 2023 report).

Getting distracted by other functional groups in a big molecule. Find the C–Cl and the N–H that react (Oct 2021 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Amide hydrolysis**: amides (and peptide links) are broken by heating under reflux with **acid** (gives the carboxylic acid + ammonium ion) or **alkali** (gives the carboxylate ion + ammonia or amine). This is how proteins are broken into amino acids (19.11).
- **Choosing the acylating agent on safety and atom-economy grounds**: ethanoyl chloride v ethanoic anhydride (WCH15 Jan 2024; see 20.3).

SPEC 19.8 Condensation and addition polymerisation

WHAT THIS SPEC POINT ASKS

Describe (i) condensation polymerisation to make polyamides such as nylon, and proteins; (ii) addition polymerisation, including poly(propenamide) and poly(ethenol).

(i) Condensation polymers: polyamides

- A **condensation polymer** forms when monomers join and **a small molecule is lost at each link** (H_2O or HCl).
- Each monomer needs **two reactive groups**, one at each end, so the chain can keep growing.
- **Polyamides** contain the amide link **–CONH–**.

Nylon-6,6

- Monomers: **hexane-1,6-diamine** $H_2N(CH_2)_6NH_2$ and **hexanedioic acid** $HOOC(CH_2)_4COOH$. The "6,6" counts the carbon atoms in each monomer.
- Each –COOH reacts with an $–NH_2$, losing **H_2O** and forming an amide link:
  $n\ H_2N(CH_2)_6NH_2 + n\ HOOC(CH_2)_4COOH \rightarrow –[NH(CH_2)_6NHCO(CH_2)_4CO]_n– + 2n\ H_2O$
- In the lab, the **diacyl chloride** (hexanedioyl dichloride) is used instead of the acid. It reacts quickly at room temperature, losing **HCl**: the "nylon rope trick", where nylon is pulled from the boundary between the two solutions.
- **Nylon-6,10** uses decanedioyl dichloride (10 C) with hexane-1,6-diamine.
- **Kevlar**: benzene-1,4-diamine + benzene-1,4-dicarboxylic acid (or its dichloride). It is a polyamide with benzene rings in the chain.

Proteins

- Proteins are natural **polyamides**. Their monomers are **amino acids**, and each amino acid has both an $–NH_2$ and a –COOH, so **one type of monomer** is enough.
- The amide link between amino acids is called a **peptide bond (–CONH–)**. Each one forms with the loss of **H_2O** (19.11).

(ii) Addition polymers

- An **addition polymer** forms when the **C=C double bonds** of many monomers open and join. **Nothing is lost**: the polymer is the only product (100% atom economy).

Poly(propenamide)

- Monomer: **propenamide** (prop-2-enamide), $CH_2=CHCONH_2$.
- $n\ CH_2=CHCONH_2 \rightarrow –[CH_2–CH(CONH_2)]_n–$

- The many –$CONH_2$ groups **hydrogen-bond with water**, so the polymer **absorbs a lot of water** to form a gel. It is used in water treatment (to clump fine particles together), as a thickener and in gels for separating proteins and DNA. The monomer is toxic; the polymer is not.

Poly(ethenol) (polyvinyl alcohol)

- The "monomer" ethenol, CH_2=CHOH, **doesn't exist** in a usable form: it rearranges at once to ethanal.
- So poly(ethenol) is made in two steps:
  1. **Ethenyl ethanoate** (CH_2=CHOOCCH_3) is addition-polymerised to **poly(ethenyl ethanoate)**.
  2. The ester groups are swapped for –OH groups by reaction with **methanol** (a hydrolysis-type exchange). How many are swapped controls how soluble the product is (19.10).
- Repeat unit: –[CH_2–CH(OH)]–

	Addition	Condensation
Monomers	C=C in each monomer	two reactive groups per monomer (e.g. –NH_2 and –COOH)
By-product	none	small molecule (H_2O or HCl) at each link
Main chain	only carbon atoms	contains other atoms (N in polyamides, O in polyesters)
Examples	poly(propenamide), poly(ethenol), poly(ethene)	nylon-6,6, Kevlar, proteins

MARK-SCHEME WORDING

- Condensation polymer: "a very large molecule / formed from a large number of monomers" **and** "a small molecule (H_2O / HCl) splits off" at each link. "Formed by addition" is not accepted (WCH15 Jun 2023).
- Nylon-6,6 monomers: "hexanedioyl dichloride" (or hexanedioic acid) and "hexane-1,6-diamine". Wrong connectivity (e.g. OH–C instead of C–OH) is penalised once.

Often asked as: identify the type of polymerisation or the monomers (MCQ); define a condensation polymer; draw the monomers of a given polymer.

COMMON MISTAKE

Drawing a monomer for an addition polymer **without the C=C**, or with methyl groups where the continuation bonds were (Jan 2022, Jun 2023 reports).

Giving the acid when the question's reaction loses HCl (it needs the acyl chloride), or the other way round.

Saying poly(ethenol) is made from ethenol.

⚠ Beyond the spec · not part of the Edexcel IAL specification

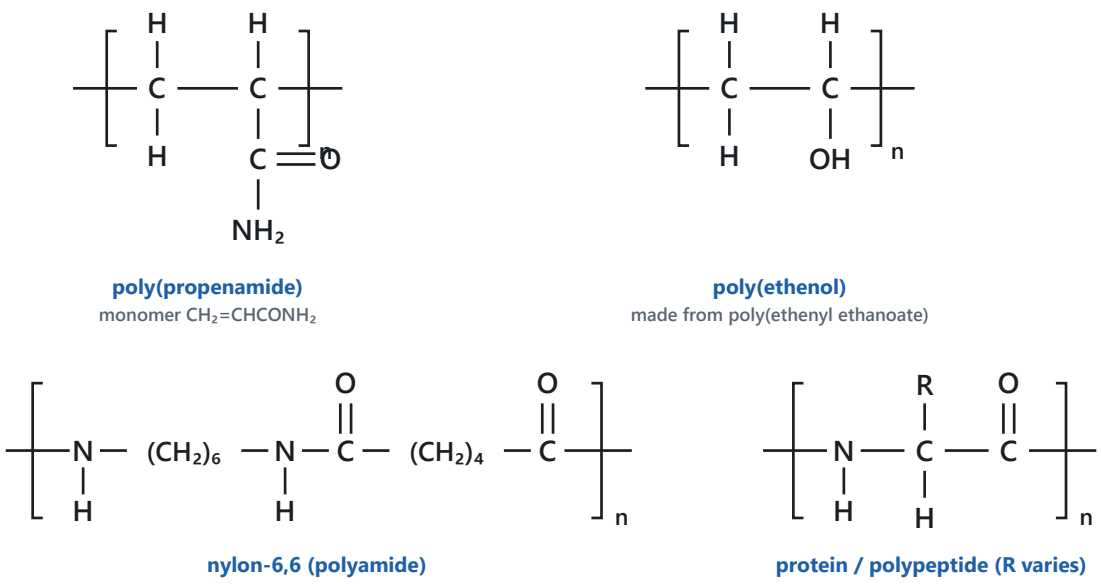
Examined in past papers even though the spec doesn't list it.

- **Nylon-6** is made by **ring-opening** polymerisation of caprolactam (a cyclic amide). The ring opens at the C–N bond. Repeat unit –[CO(CH₂)₅NH]– (WCH15 Jun 2023).
- **Hydrogels**: poly(propenamide) (PAM), poly(propenoic acid) (PAA) and PVP absorb water. PAA releases a drug above pH 8, because its –COOH groups become –COO⁻ and the negative charges **repel** (WCH15 Jun 2023).

SPEC 19.9 Repeat units of these polymers

WHAT THIS SPEC POINT ASKS

Draw the structural formulae of the repeat units of the polymers in 19.8.



Repeat units. The continuation bonds go out through the brackets, and n follows the closing bracket.

Polymer	Repeat unit
poly(propenamide)	–[CH₂–CH(CONH₂)]–
poly(ethenol)	–[CH₂–CH(OH)]–
nylon-6,6	–[NH(CH₂)₆NH–CO(CH₂)₄CO]–

Polymer	Repeat unit
Kevlar	–[NH–C_6H_4–NH–CO–C_6H_4–CO]– (rings joined at positions 1 and 4)
polypeptide (general)	–[NH–CHR–CO]–
poly(glycine)	–[NH–CH_2–CO]–

Rules

- **Addition polymer**: the repeat unit is the monomer with its C=C turned into C–C and **two continuation bonds**, one from each of the two former C=C carbons. Substituents hang off the chain.
- **Condensation polymer from two monomers**: the repeat unit contains **one of each monomer** joined by one amide link, with an NH at one end and a CO at the other.
- **Continuation bonds** cross the brackets. Don't put CH_3 or H at the ends.

Working backwards: monomers from a polymer

- **Addition**: take the two main-chain carbons of the repeat unit, put a **C=C** between them, and remove the continuation bonds.
- **Polyamide**: break each **C–N bond of the amide link**. Add **OH** to the C=O end (acid) or **Cl** (acyl chloride), and **H** to the N end (amine).

WORKED EXAMPLE

Worked example: degree of polymerisation

A sample of nylon-6,6 has a mean relative molecular mass of 45 000. How many repeat units are there in an average chain?

Repeat unit: $C_{12}H_{22}N_2O_2$.

M_r = (12 × 12.0) + (22 × 1.0) + (2 × 14.0) + (2 × 16.0) = 144 + 22 + 28 + 32 = **226**.

Number of repeat units = 45 000 ÷ 226 = **about 200**.

WORKED EXAMPLE

Worked example: percentage by mass of OH in poly(ethenol)

Repeat unit $CH_2CH(OH)$ = C_2H_4O, M_r = 44.0. Mass of OH in it = 17.0.

% OH = 17.0 ÷ 44.0 × 100 = **38.6%**.

The answer is the same for any number of units: for three units, 51 ÷ 132 = 38.6% (WCH15 Jun 2024).

MARK-SCHEME WORDING

- Repeat unit with "continuation bonds" and the amide link correct, e.g. "–$CO(CH_2)_5NHCO(CH_2)_5NH$–" for two units of nylon-6.
- Number of repeat units = polymer M_r ÷ repeat-unit M_r (e.g. 300 000 ÷ 158 = 1900, WCH15 Jan 2022). Number of molecules = moles × 6.02×10^{23} (WCH15 Jun 2023).

Often asked as: draw a repeat unit, give the monomers, calculate the repeat-unit M_r and the number of units.

COMMON MISTAKE

Miscounting the monomer's M_r (extra carbons at the continuation bonds; Jun 2023 report).

Missing the H on the N in a skeletal or displayed polyamide.

Joining a cyclic monomer through the wrong atom (e.g. caprolactam rings joined through the N; Jun 2023 report).

SPEC 19.10 Physical properties of polyamides and poly(ethenol)

WHAT THIS SPEC POINT ASKS

Comment on the physical properties of polyamides, and on the solubility of poly(ethenol) in water, in terms of hydrogen bonding. Include soluble laundry bags and liquid-detergent capsules.

Polyamides

- The **N–H** of one chain forms **hydrogen bonds** with the **C=O** of a neighbouring chain. There are also London forces along the chains.
- These interchain forces are **much stronger than the London forces alone** between poly(alkene) chains. So polyamides have **higher melting temperatures** and **higher tensile strength**, and make good **fibres** (clothing, ropes, carpets).
- **Kevlar**: flat, rigid chains (benzene rings in the backbone) line up regularly, with many hydrogen bonds between them. This makes it very strong and heat-resistant: used in **bullet-proof vests**, protective gloves and helmets.
- Polyamides can be **hydrolysed** at the amide links by acids or alkalis, which weakens them.

Poly(ethenol)

- Every second carbon carries an **–OH group**. These form **hydrogen bonds with water** molecules, so poly(ethenol) **dissolves in water**.
- Solubility depends on **how many OH groups** there are:

- Leaving some ester groups in place (partial hydrolysis) gives a polymer that dissolves in cold or warm water.
- Almost fully converted poly(ethenol) has so many hydrogen bonds **between its own chains** that it dissolves only in **hot** water.

- **Uses**:
  - **Soluble laundry bags** in hospitals: dirty linen goes into the washing machine still in the bag, which dissolves in the wash. Staff don't handle the linen, so there is **no risk of cross-contamination** (infection).
  - **Liquid-detergent capsules** (liquitabs): the film holds the liquid, which contains very little water, and dissolves when it meets the wash water.
  - Others: dissolving stitches and water-soluble packaging.

MARK-SCHEME WORDING

- Polyamide v polyalkene melting point: "hydrogen bonding (and London forces) between polyamide chains is stronger than the London forces between polyalkene chains, so more energy is needed to separate the chains". Any mention of covalent bonds breaking scores 0. Hydrogen bonding "to water" is not accepted here (WCH15 Oct 2023).
- Poly(ethenol): "OH groups can form hydrogen bonds with water", "high percentage of OH groups, so soluble" (WCH15 Jun 2024).
- PAM absorbs water: "can form many hydrogen bonds with water", "at the NH_2 and the C=O" (both sites), plus a correct hydrogen-bond diagram (WCH15 Jun 2023).
- Laundry bags: "no risk of cross-contamination", "no need to wash (and dry) the bags". "Biodegradable", cost and disposal are ignored.

Often asked as: explain a melting temperature or a solubility; explain the advantage of soluble laundry bags; draw hydrogen bonds between a polymer and water.

COMMON MISTAKE

Saying the amide **bond** is strong. Melting a polymer separates chains: it breaks **intermolecular** forces, not covalent bonds (Oct 2023 report).

Writing "biodegradable" as the advantage of laundry bags (Jun 2024 report).

Showing hydrogen bonds at only one site of the $–CONH_2$ group (Jun 2023 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Plasticisers: glycerol (propane-1,2,3-triol) is added to poly(ethenol) film to make it **more flexible**. "Increases solubility" is not accepted (WCH15 Jun 2024). That paper also included a calculation of polymer density and plasticiser percentage.

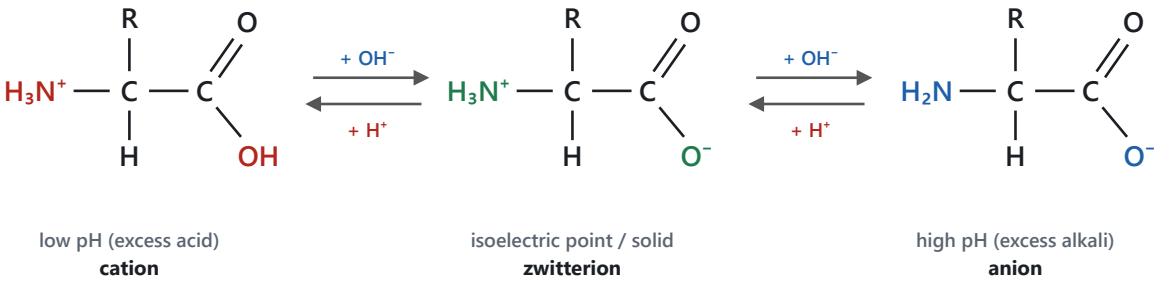
SPEC 19.11 Investigating the behaviour of amino acids

WHAT THIS SPEC POINT ASKS

Describe experiments showing (i) the acid–base behaviour of amino acids and zwitterion formation; (ii) the effect of their aqueous solutions on plane-polarised monochromatic light; (iii) peptide bond formation by condensation.

(i) Acid, base and zwitterion

- An amino acid has a basic group (–NH_2) and an acidic group (–COOH) in the same molecule. The –COOH can pass its H^+ to the –NH_2 **inside the molecule**, giving a **zwitterion**: $^+$**H$_3$N–CHR–COO$^-$**. It has both charges but **no overall charge**.
- In the **solid**, amino acids exist as zwitterions. The ions attract each other strongly (like an ionic lattice), so amino acids:
  - have **high melting temperatures** (often over 200 °C; many decompose instead of melting);
  - **dissolve in water** but not in non-polar solvents.
- Amino acids are **amphoteric**: they react with both acids and bases.



The form depends on pH. Adding acid protonates the COO^-; adding alkali removes H^+ from the NH_3^+.

Conditions	Form	Formula (glycine)
Acid (low pH)	**cation**	$^+H_3NCH_2COOH$
Isoelectric point (and the solid)	**zwitterion**	$^+H_3NCH_2COO^-$
Alkali (high pH)	**anion**	$H_2NCH_2COO^-$

Equations (glycine):

- with acid: $^{+}H_3NCH_2COO^- + H^+ \rightarrow {}^{+}H_3NCH_2COOH$
- with alkali: $^{+}H_3NCH_2COO^- + OH^- \rightarrow H_2NCH_2COO^- + H_2O$

Experiments

- Measure the pH of a glycine solution with a **pH meter**: close to neutral (about 6), because the zwitterion has both an acidic group and a basic group.
- Add a few drops of dilute HCl, then of dilute NaOH, to glycine solution, and to the same volume of water. The pH of the glycine solution changes **much less**, because the zwitterion **reacts with both H^+ and OH^-** (it acts as a buffer).
- Compare melting temperatures, or solubility in water and in hexane, with a similar-sized molecule that isn't ionic.

(ii) Plane-polarised light (optical activity)

- Every amino acid except **glycine** has a **chiral carbon**: carbon 2 is bonded to **four different groups** (H, NH_2, COOH and R). So it has two **optical isomers** (enantiomers): non-superimposable mirror images.
- A solution of **one** enantiomer **rotates the plane of plane-polarised light**. Its mirror image rotates it by the **same angle** in the **opposite direction**.
- **Experiment (polarimeter)**:
  1. Shine **monochromatic light** (one wavelength, e.g. a sodium lamp) through a **polariser**, giving plane-polarised light.
  2. Pass it through a tube of the amino acid solution.
  3. Turn an **analyser** (a second polariser) until the light is brightest (or darkest). Compare the angle with a tube of pure water.
  4. The difference is the **angle of rotation**: clockwise (+) or anticlockwise (−).
- Monochromatic light is used because the angle of rotation depends on the wavelength.
- **Results**: natural amino acids (from proteins) are **single enantiomers**, so they rotate the light. **Glycine** has no chiral carbon, so no rotation. An amino acid made in the lab is usually a **racemic mixture** (equal amounts of both enantiomers), so the rotations cancel and there is **no overall rotation**.

(iii) Peptide bonds

- The –COOH of one amino acid reacts with the $–NH_2$ of another. **Water is lost** and a **peptide bond, –CONH–**, forms: a **condensation** reaction.

$$H_2NCH_2COOH + H_2NCH(CH_3)COOH \rightarrow H_2NCH_2CONHCH(CH_3)COOH + H_2O$$

- Two **different** amino acids can join in **two ways** (Gly–Ala and Ala–Gly), so there are two possible dipeptides. Each still has a free $–NH_2$ at one end and a free –COOH at the other, so more amino acids can join: **polypeptides** and **proteins**.
- **Experiment (reverse direction)**: **hydrolyse** a protein or dipeptide by heating under reflux with **6 mol dm^{-3} HCl** for several hours. This breaks the peptide bonds. Then identify the amino acids by **paper or thin-layer**

chromatography: spray with **ninhydrin**, which shows the spots as purple, and compare the **R_f values** with known amino acids.

- On hydrolysis, H goes back on the N and OH goes back on the C=O at each peptide bond.

WORKED EXAMPLE

Worked example: hydrolysis and number of molecules (aspartame)

Aspartame ($C_{14}H_{18}N_2O_5$, M_r 294) is the methyl ester of a dipeptide of aspartic acid and phenylalanine.

Full hydrolysis breaks the peptide bond **and** the ester link. Products: **2-aminobutanedioic acid** (aspartic acid), **2-amino-3-phenylpropanoic acid** (phenylalanine) and **methanol**.

A drink contains 200 mg of aspartame. How many molecules?

Moles = 0.200 g ÷ 294 g mol^{-1} = 6.80×10^{-4} mol.

Molecules = $6.80 \times 10^{-4} \times 6.02 \times 10^{23}$ = **4.10×10^{20}**.

Convert mg to g first: this conversion is the usual slip (WCH15 Jan 2023).

MARK-SCHEME WORDING

- Zwitterion: "+ anywhere on the NH_3 group"; carboxylate drawn as COO^- (or with the charge delocalised). Leaving the amine as NH_2 loses the mark.
- Explaining zwitterion formation: "amine group can accept a proton / become $–NH_3^+$" and "carboxylic acid group can lose a proton / become $–CO_2^-$" (WCH15 Oct 2025).
- Chiral carbon: "attached to four different **groups**" ("four different molecules" is not accepted); a mirror-image pair of 3D diagrams scores both marks.
- Hydrolysis products: drawn with no trailing bonds; NH_3^+ accepted for NH_2 (WCH15 Jan 2023).

Often asked as: draw the zwitterion, cation or anion; draw a dipeptide and circle the peptide bond; identify chiral carbons; give the hydrolysis products; MCQs on pH and form.

COMMON MISTAKE

Drawing the "zwitterion" with an unchanged NH_2 (Jan 2023 report).

Reading a dipeptide as a polymer repeat unit, or cutting hydrolysis products into fragments with dangling bonds (Jan 2023 report).

Saying a chiral molecule has cis–trans isomers, or that a C=C carbon can be chiral (Oct 2022 report).

Saying the racemic mixture "doesn't rotate light because it isn't chiral". Each molecule is chiral; the **equal amounts** of opposite rotations cancel.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Chromatography**: R_f = distance moved by the spot ÷ distance moved by the solvent, both measured **from the baseline** (WCH15 Jan 2023).
- **Optical isomerism outside amino acids** (coniine, gingerol, metal complexes): the same "four different groups" rule. Wedge and dash bonds show the 3D arrangement (WCH15 Oct 2022, Jan 2024).

SPEC 19.12 Core Practical 15: analysis of inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Core Practical 15: identify unknown inorganic and organic substances using test-tube reactions.

How to approach it

- Work from **observations** to **conclusions**: each test either confirms or rules out a group or ion.
- Record **exactly what you see**, with colours **before and after** ("orange solution turns green"), and whether a gas, precipitate or solution forms.
- Combine evidence: one test rarely proves a structure. Data such as formula and spectra (20.1) finish the job.

CORE PRACTICAL

Core practical 15: approach

1. **Cations**: add NaOH(aq) drop by drop, then in excess; repeat with NH_3(aq). Note precipitate colours and whether they dissolve in excess (full colour table in 17.22). Flame tests for Group 1 and 2 ions.
2. **Anions**: test for carbonate, sulfate and halides (table below).
3. **Organic**: test for each functional group in turn (table below), using small samples and a fresh sample for each test.
4. Warm only with a **water bath** (most organic liquids are flammable). Work in a fume cupboard for volatile, toxic or corrosive reagents, e.g. PCl_5 and bromine.

Key inorganic tests

Ion	Test	Positive result
CO_3^{2-}	dilute acid; gas through limewater	fizzing; limewater turns cloudy (CO_2)

Ion	Test	Positive result
SO_4^{2-}	acidify with HCl, add $BaCl_2$(aq)	white precipitate ($BaSO_4$)
Cl^- / Br^- / I^-	acidify with HNO_3, add $AgNO_3$(aq)	white / cream / yellow precipitate. Then dilute NH_3 dissolves AgCl, conc. NH_3 dissolves AgBr, AgI doesn't dissolve
NH_4^+	warm with NaOH(aq)	gas turns damp red litmus blue (NH_3)
Cu^{2+}	NaOH(aq); NH_3(aq)	blue precipitate; with excess NH_3, deep blue solution
Fe^{2+}	NaOH(aq)	green precipitate, turns orange-brown at the surface in air
Fe^{3+}	NaOH(aq)	orange-brown precipitate
Cr^{3+}	NaOH(aq)	grey-green precipitate; dissolves in excess to a dark green solution

Key organic tests

Group	Test	Positive result
C=C (alkene)	bromine water, shake	orange → **colourless**
arene	burn a small sample	**very smoky (sooty)** flame
–OH (alcohol, acid)	PCl_5 (dry)	**steamy fumes** of HCl
1° or 2° alcohol, aldehyde	acidified $K_2Cr_2O_7$, warm	**orange → green** (no change with 3° alcohols or ketones)
C=O (aldehyde or ketone)	2,4-dinitrophenylhydrazine	**orange (or yellow) precipitate**
aldehyde (not ketone)	Tollens' reagent, warm	**silver mirror**
aldehyde (not ketone)	Fehling's or Benedict's solution, warm	blue solution → **red precipitate**
CH_3CO– or $CH_3CH(OH)$–	iodine + NaOH(aq), warm (iodoform)	**pale yellow precipitate** of CHI_3 (antiseptic smell)
–COOH	Na_2CO_3 or $NaHCO_3$ solution	**fizzing**; CO_2 turns limewater cloudy

Group	Test	Positive result
halogenoalkane	warm with NaOH(aq), acidify with HNO_3, add $AgNO_3$	white / cream / yellow precipitate shows Cl / Br / I
phenol	bromine water	decolourises; **white precipitate**
amine	damp red litmus; smell; Cu^{2+}(aq)	litmus blue; fishy smell; pale blue precipitate, then deep blue solution with excess (butylamine)

MARK-SCHEME WORDING

- Unknown deduction (WCH15 Jun 2022, 6 marks): "sooty flame indicates benzene ring / arene / high C : H ratio"; "orange precipitate with 2,4-DNPH indicates carbonyl"; "silver precipitate with Tollens' indicates aldehyde"; "decolourises bromine water indicates C=C"; "exists as geometric isomers indicates only 1 H on each carbon of the C=C".
- Alcohol class: "does not react with acidified potassium dichromate(VI), so tertiary alcohol" (WCH15 Oct 2020).
- Iodoform: CHI_3 alone scores for $CH_3CO–$ or $CH_3CH(OH)–$ (WCH15 Jan 2023).
- Inorganic: "pale green" solution means **iron(II)**; then use the masses to show it is a hydrate (WCH15 Oct 2021).

Often asked as: deduce a structure from a set of test results plus formula data; give the reagent and the observation for a named group; an inorganic salt identified from tests and percentage composition.

COMMON MISTAKE

Drawing a structure that contradicts one of the results you were given, e.g. a primary alcohol when dichromate stayed orange (Oct 2020 report).

Giving only the final colour. Write "orange to green", not "green".

Not linking a pale green solution to Fe^{2+}, or missing the water of crystallisation (Oct 2021 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Cerium(IV) ammonium nitrate** turns from yellow to **red** with an alcohol (WCH15 Oct 2020).
- **Iron(III) chloride** solution gives a **purple** colour with phenols. It is useful for checking aspirin for unreacted 2-hydroxybenzoic acid (20.4).

Quick checks

QUICK CHECK

Which is the strongest base?

- ammonia
- phenylamine
- diethylamine
- ethanamide

Two electron-releasing ethyl groups make diethylamine's lone pair the most available. Phenylamine's lone pair is delocalised into the ring and ethanamide's towards the C=O, so both are much weaker.

QUICK CHECK

Why is a large excess of ammonia used to make ethylamine from bromoethane?

- ammonia is a catalyst
- so that bromoethane is more likely to meet ammonia than the ethylamine already formed, reducing further substitution
- to neutralise the HBr formed
- ammonia is the solvent

Ethylamine is itself a nucleophile and can react with more bromoethane to give secondary and tertiary amines. Swamping it with ammonia makes the primary amine the main product.

QUICK CHECK

Why must the diazotisation of phenylamine be kept below 10 °C?

- nitrous acid boils at 10 °C
- the azo dye decomposes
- above 10 °C the diazonium ion reacts with water to form phenol and nitrogen gas
- phenylamine freezes at higher temperatures

Benzenediazonium ions are unstable when warm. The low temperature keeps them long enough to couple with phenol.

QUICK CHECK

How many molecules of water are released when 50 molecules of hexanedioic acid react with 50 molecules of hexane-1,6-diamine to form one nylon 6,6 chain?

- 50
- 99
- 100
- 101

100 monomers joined in a single chain need 99 amide links, each releasing one water molecule. (For large n, 2n − 1 is usually written as 2n.)

QUICK CHECK

Why do amino acids have high melting points?

- they are large molecules with strong London forces
- they form covalent networks
- they exist as zwitterions, held together by strong ionic attractions
- they hydrogen bond more than water

In the solid, each molecule carries NH_3^+ and COO^- groups, so the crystal is held together like an ionic lattice.

QUICK CHECK

Which form of glycine is present at pH 1?

- $H_2NCH_2COO^-$
- H_2NCH_2COOH
- $^+H_3NCH_2COOH$
- $^+H_3NCH_2COO^-$

In strong acid both groups are protonated, giving the cation. H_2NCH_2COOH, with no charges, is never the main form in solution.

QUICK CHECK

Why do polyamides usually have higher melting points than polyesters of similar chain length?

- polyamides have stronger covalent bonds in the chain
- polyamide chains hydrogen bond to each other through N–H and C=O groups
- polyesters have no permanent dipoles
- polyamides are addition polymers

Both have C=O dipoles, but only polyamides have N–H groups able to hydrogen bond to C=O groups in neighbouring chains.

QUICK CHECK

Glycine (G) and alanine (A) are joined by a single peptide link. How many different dipeptides can form from one of each?

- 1
- 2
- 3
- 4

Gly–Ala and Ala–Gly are different: either amino acid can provide the –COOH that forms the peptide link. (Allowing Gly–Gly and Ala–Ala as well would give four.)